

AFL Data Dictionary & Target Definitions

Week 6, Day 1 — AFL Data Foundations. Contract for every model built this week; if a target or feature definition changes, it changes here first.

1. Source Tables

Table	Grain (1 row =)	Join Key
afl_players_round_by_round_stats_raw.csv	One player, one match	player_id → players_info.id
team_matches_home_away_raw.csv	One team, one match (2 rows/match)	(team,opponent,date,round) → match_id
players_info.csv	One player (career bio)	id
seasonal_stats.csv	One player, one team, one season	player_id + year

2. Prediction Targets

Target	Definition	Level
target_result	HOME_WIN / AWAY_WIN / DRAW, from home perspective	match
target_margin	home_score – away_score (signed)	match
disposals	kicks + handballs	player-match
goals	goals kicked	player-match
fantasy_points	3×kicks + 2×handballs + 3×marks + 4×tackles + FF – 3×FA + 6×goals + behinds	player-match
brownlow_votes	Official best-afield votes (0/1/2/3)	player-match

Match winner is framed as 3-class classification (not binary) because the draw rate is ~0.82% — rare enough to keep as its own class rather than silently folding into win/loss, which would bias the model. Margin is kept as a secondary regression target for confidence/ranking use cases.

3. Feature Dictionary (v1)

Feature	Description	Window
form_win_rate_last5	Win rate, team's last 5 matches	rolling 5, shift(1)
form_avg_score_last5	Avg points scored, last 5 matches	rolling 5, shift(1)
form_avg_margin_last5	Avg signed margin, last 5 matches	rolling 5, shift(1)
win_streak	Signed streak (+wins / –losses) coming in	all history, shift(1)
days_rest	Days since team's previous match	1-lag diff
ladder_pos_before	Ladder rank within season, before this round	season-cumulative, shift(1)
h2h_win_rate_prior	Win rate vs. this opponent, all prior meetings	full history, strictly prior
is_interstate	1 if team is playing outside its home state	static team/venue → state map
player_avg_{disposals,goals,fantasy_points}_last5	Player's rolling average, last 5 games	rolling 5, shift(1)

Leakage rule: every rolling/cumulative feature applies shift(1) before windowing, so a match's own outcome is never visible in its own feature row. Verified two ways in the notebook: (1) each team's first-ever match has 100% null form/streak features, and (2) a manual recomputation using only strictly-earlier matches is asserted equal to the pipeline output.

4. Data Quality Issues Found & Handled

- Team names: 40 raw variants of 20 real teams in team_matches (tabs/whitespace, "W. Bulldogs" vs "Western Bulldogs", 2,345 lowercase rows in the opponent column) — fixed via strip + title-case + explicit alias map.
- Venue names: 469 rows with trailing \n — stripped.

- 10 exact duplicate rows in player-match data — dropped.
- rounds.score column is 100% null — dropped as dead.
- 722 rows with impossible negative disposals — set to null, not zero-filled.
- Advanced stats (hit_outs, contested_possessions) are structurally missing pre-2000 — the stat wasn't tracked yet; left as null, not imputed.
- No match_id or position column exists in source data — match_id constructed and validated (100% clean 2-row pairing across 7,904 matches); position approximated only as a documented heuristic proxy, not presented as ground truth.
- Club history: Fitzroy Lions and Brisbane Bears are distinct historical entities from Brisbane Lions (post-1997 merger) — kept separate, not collapsed.
- Bug caught in review: ladder_pos_before was initially ranking each team only against its own match opponent (grouped by match_id, always 2 rows) instead of the full competition — fixed to rank within (year, round) across all teams; verified against the 2019 season (range now correctly 1–18, the real team count).

5. Train / Holdout Split

Strict time-based split: train on all seasons except the most recent, test on the held-out season(s). A random split leaks because ladder position, form, and head-to-head are season-accumulating — a random split can put an early-round match in test while a later round from the same season (which used information building on it) sits in training. Implemented once as `time_based_split()` and reused by every model this week for comparable, leak-free results.

Realistic ceiling: always-pick-home already gets ~59% from home ground advantage alone — that's the floor, not 50%. Published AFL prediction models typically land 65–72% on match winner. Sports outcomes carry genuine irreducible randomness (injuries, umpiring, weather); a holdout accuracy near 90%+ should be treated as a leakage red flag, not a result to celebrate.